

(a) Can you set up a simulation for cube particles? 

I'll set up a cube particle simulation. I need a few details to configure it:

- Dimension**: Cubes are 3D particles (confirmed from capabilities).
- Ensemble**: NVT (fixed volume) or NPT (fixed pressure)?
- Total particle number**: How many cubes? (e.g., 512, 1000, 2048)

If you'd like sensible defaults, I can proceed with:
`{`example parameters`}`
 Shall I go ahead with these defaults, or do you want different values?

let's try 300 particles in NPT ensemble 

Simulation problem created at `{`working_dir`}` **Configuration**:

- Dimension: 3D - Ensemble: NPT - Particles: 300 cubes
- Default pressure `P``: 1.0 - Default `sample_steps``: 20,000

Next step: planning runs. Would you like to: `{`suggested runs`}`
 [create simulation set up file]

let's use particle side 1, let's sample $5e5$ steps and sweep 5 pressure from 2 to 10 

Planned 5 runs: `{`detail about the planned runs`}`
 Ready to launch the simulations? `{`more related information`}`
 [create simulation planning files and folders]

go ahead 

[launch simulations, run in parallel]
 Simulations launched as background job ``wf_fd****``.
Monitor progress: `{`simulation progress`}`

analyze the trajectory data when it's finished 

[call data analysis tool]
 Each run directory contains: `{`simulation output file detail`}`
 Let me know if you want me to inspect specific results...

